

Rethinking Noisy Label Learning in Real-world Annotation Scenarios from the Noise-type Perspective

Renyu Zhu¹, Haoyu Liu¹, Runze Wu¹, Minmin Lin¹, Tangjie Lv¹, Changjie Fan¹, Haobo Wang²

¹NetEase Fuxi AI Lab, Hangzhou, Zhejiang, China

{zhurenryu, liuhaoyu03, wurunze1, linminmin01, hzlv tangjie, fanchangjie}@corp.netease.com

²Zhejiang Univeristy, Hangzhou, Zhejiang, China

wanghaobo@zju.edu.cn

Abstract

In this paper, we investigate the problem of learning with noisy labels in real-world annotation scenarios, where noise can be categorized into two types: factual noise and ambiguous noise. To better distinguish these noise types and utilize their semantics, we propose a novel sample selection-based approach for noisy label learning, called Proto-semi. Proto-semi initially divides all samples into the confident and unconfident datasets via warm-up. By leveraging the confident dataset, prototype vectors are constructed to capture class characteristics. Subsequently, the distances between the unconfident samples and the prototype vectors are calculated to facilitate noise classification. Based on these distances, the labels are either corrected or retained, resulting in the refinement of the confident and unconfident datasets. Finally, we introduce a semi-supervised learning method to enhance training. Empirical evaluations on a real-world annotated dataset substantiate the robustness of Proto-semi in handling the problem of learning from noisy labels. Meanwhile, the prototype-based repartitioning strategy is shown to be effective in mitigating the adverse impact of label noise. Our code and data are available at <https://anonymous.4open.science/r/ProtoSemi-C61F>.

Introduction

The rapid development of deep learning technologies such as ChatGPT (?) is inseparable from the support of high-quality labeled data. Unfortunately, creating such data is costly as it requires expert annotation. To address this issue, researchers have introduced low-cost data collection methods such as crowdsourcing (?). However, the quality of the collected data is hard to fully guarantee, resulting in the production of noisy labels.

The occurrence of noisy labels may cause the model to suffer from the memorization phenomenon (??), which leads to overfitting and performance decrease. As a result, numerous works (??) have been devoted to learning from noisy labels, such as robust structures (??), robust regularization (??), robust loss functions (??), and sample selection-based methods (??). Among them, the sample selection-based methods first identify clean samples with true labels from all samples, and introduce semi-supervised learning (?) to enhance training, which achieve state-of-the-art performance in noisy label learning.

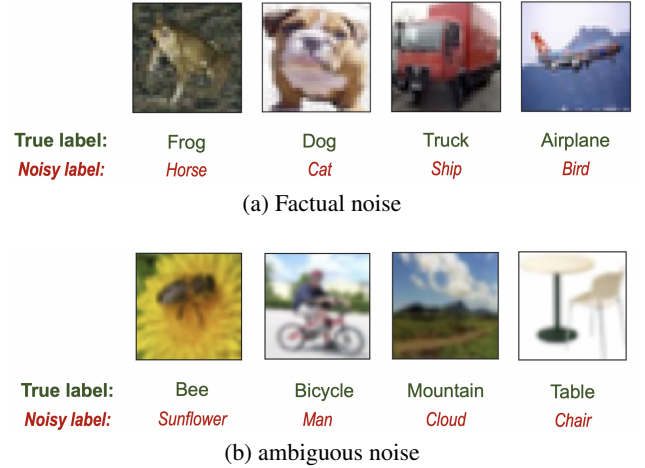


Figure 1: Examples of the two noise types in real-world annotation scenarios.

Meanwhile, previous works (??) have indicated that noise generated during annotation in real-world scenarios can be divided into two types: 1) **Factual noise**: the annotator mistakenly labels the sample due to carelessness or lack of knowledge during the annotation process; 2) **ambiguous noise**: the annotator labels the sample incorrectly due to its ambiguity, such as multiple labels. The examples of the two noise types are shown in Figure 1. However, previous sample selection-based methods (??) neglect the noise types and simply remove the labels of all noisy samples. It inspires us to rethink noisy label learning in real-world annotation scenarios: *How to distinguish the noise types and utilize their semantics?*

To answer the question, in this paper, we propose a novel sample selection-based method, called Proto-semi, for noisy label learning in real-world annotation scenarios. Similar to previous sample selection-based methods (???), We first perform a warm-up pretraining to divide the training samples into confident and unconfident datasets. Then we create a prototype vector for each class based on the corresponding samples in the confident dataset, and calculate the distance between each sample in the unconfident dataset and

these prototypes. After that, the noise types of samples in the unconfident dataset are identified based on the calculated distance: 1) For samples with a small distance, we consider them to be factual noise and perform label correction. 2) For samples with a distance that is neither too large nor too small, we categorize it as an ambiguous error and correct or retain the label based on the probability according to the distance value. The corrected or retained data, along with their labels, are moved to the confident dataset. Finally, we introduce the semi-supervised learning method (?) to train the model based on the refined confident and unconfident datasets. In summary, our contributions are followed as:

- We present a new noise-type perspective on noisy label learning in real-world scenarios.
- We propose a novel sample selection-based method called Proto-semi for noisy label learning. Proto-semi introduces prototypes to repartition confident and unconfident datasets based on the types of noise.
- Experiments on the CIFAR-10 (?) dataset demonstrate the effectiveness of Proto-semi for noisy label learning in real-world scenarios. Additionally, we use prototype visualization and detailed statistical analysis to confirm that Proto-semi can effectively identify the type of noise and correct the labels for factual noise.

Related Works

Learning From Noisy Labels

The existing literature on noisy label learning can be broadly classified into four categories. The first category focuses on designing more robust architectures, such as dedicated architectures (??) and the addition of noise adaptation layers (??) at the top of the softmax layer to model the noise transition matrix. The second category centers on robust regularization, including explicit (??) and implicit regularization (??), both of which aim to prevent overfitting in noisy data. The third category focuses on developing more robust loss functions, such as loss regularization (?), loss correction (?), and loss reweighing (?). Finally, the last category is sample selection-based (??), which involves selecting true labeled examples from noisy training datasets and has been shown to be the most effective method among all categories.

Sample selection-based methods can be further classified into two categories: loss-based sampling (??) and feature-based sampling (??). Loss-based sample selection methods are based on the small loss hypothesis, which assumes that noisy data tends to incur a high loss due to the model's inability to accurately classify such data. Feature-based sampling methods distinguish clean and noisy samples by leveraging features. However, both of these categories simply categorize all predicted noisy samples as unlabeled, which does not fully exploit the training data. Therefore, in this paper, we propose a novel noise-type-based sample selection method to enhance the utilization of annotation information in training data.

Prototype Neural Networks

Prototype-based metric learning methods (?) have emerged as a promising technique in numerous research domains, in-

cluding few-shot learning (?) and meta-learning (?). ProtoNet (?), which introduces prototypes into deep learning, is a seminal work in this area. Subsequent studies (??) have focused on improving prototype estimation to enhance classification performance. The success of prototype-based models suggests that prototypes can serve as representative embeddings of instances from the same classes, thereby inspiring us to explore the use of prototype networks in the context of noisy label learning.

Problem Definition

In practical problems, it is often difficult to obtain samples from the true distribution D of a pair of random variables $(\mathbf{X}, \mathbf{Y}) \in \mathcal{X} \times \{1, K\}$ due to random label corruption. Here, $\mathbf{X}$ is the sample variable, $\mathbf{Y}$ is the label variable, $\mathcal{X}$ represents the feature space, and K is the number of classes. Instead, in the context of noisy label learning, we can only access a set of samples $\{(\mathbf{x}_i, \tilde{y}_i)\}_{i=1}^n$ that are independently drawn from the distribution $\tilde{D}$ of noisy samples $(\mathbf{X}, \tilde{\mathbf{Y}})$, where $\tilde{\mathbf{Y}}$ represents the noisy label variable. The objective is to learn a robust classifier based on noisy samples that can accurately classify test samples.

Method

In this section, we introduce our proposed method Proto-semi for learning with noisy labels in real-world annotation scenarios. Proto-semi is mainly composed of two steps: sample selection and semi-supervised learning. Sample selection is used to divide the whole dataset into the confident and unconfident datasets, in which we propose a prototype-based repartitioning strategy to identify the types of noise and correct their labels. And then semi-supervised learning is introduced to enhance the training based on the repartitioned confident and unconfident datasets. Details of the two steps are described in the following.

Sample Selection

Similar to previous label-noise learning methods based on sample selection (???), we begin by dividing the original noisy dataset $\{(\mathbf{x}_i, \tilde{y}_i)\}_{i=1}^n$ into two distinct subsets: a labeled confident dataset $\mathcal{D}_l$ and an unlabeled unconfident dataset $\mathcal{D}_u$. In this work, we accomplish this through a warm-up phase, during which we optimize a training function as follows:

$$\min_{\Theta} \frac{1}{n} \sum_{i=1}^n \mathcal{L}(f(\mathbf{x}_i; \Theta), \tilde{y}_i), \quad (1)$$

where $f(\cdot; \Theta)$ denotes the classifier, such as deep neural networks, Θ is the parameters of classifier, and $\mathcal{L}(\cdot)$ is the cross-entropy loss function. To avoid the memorization phenomenon (?), we only perform a small number of epochs in the warm-up phase. Upon completing the warm-up phase, we proceed to identify samples across all examples whose classifier soft outputs are consistent with their current labels. The identified samples are put into the confident dataset $\mathcal{D}_l$ and their labels are retained. In contrast, samples whose soft outputs and current labels are inconsistent are dropped

into unconfident dataset $\mathcal{D}_u$ and their labels are discarded. In short, the split process can be summarized by the following two formulas:

$$\mathcal{D}_l = \{(\mathbf{x}_i, \tilde{y}_i) | \tilde{y}_i = \hat{y}_i, i = 1, \dots, n\}, \quad (2)$$

$$\mathcal{D}_u = \{\mathbf{x}_i | \tilde{y}_i \neq \hat{y}_i, i = 1, \dots, n\}, \quad (3)$$

where $\hat{y}_i$ represents the soft output of classifier $f(\cdot; \Theta)$ on sample i . As pointed out by previous works (?), during the warm-up phase, the model gradually converges to clean samples that make up the majority of all samples¹. Therefore, based on the warm-up dividing, the majority of clean samples and noisy samples are assigned to the confident dataset $\mathcal{D}_l$ and unconfident dataset $\mathcal{D}_u$, respectively. However, as aforementioned in the Introduction, noise in the human annotation scenario can be mainly divided into factual noise and ambiguous noise. To distinguish these noises and utilize the information effectively, we propose a prototype-based repartitioning strategy.

First, we construct an M -dimensional representation $\mathbf{c}_k \in \mathcal{R}^M$, or *prototype*, of each class through an embedding function. In this paper, we use the classifier after warm-up and remove its last linear layer as the embedding function, tagged as $f'(\cdot; \Theta)$. And each prototype is the mean vector of all sample embeddings belonging to its class:

$$\mathbf{c}_k = \frac{1}{|D_l^k|} \sum_{(\mathbf{x}_i, \tilde{y}_i) \in D_l^k} f'(\mathbf{x}_i; \Theta), \quad (4)$$

where D_l^k denotes the all samples that belong to the class k in the confident dataset D_l , and $|D_l^k|$ is the number of samples in D_l^k . Similarly, we can obtain the prototype vectors of classes in addition to the class k based on Eq. (4). And we aggregate all the prototype vectors of each class together to obtain the prototype matrix $\mathbf{C}$.

After that, for each sample u in the unconfident dataset $\mathcal{D}_u$, we calculate the distance between the sample and all prototypes:

$$\mathbf{d} = \text{Distance}(f'(\mathbf{x}_u; \Theta), \mathbf{C}), \quad (5)$$

where $\mathbf{d}$ denotes the distance vector and $\text{Distance}(\cdot, \cdot)$ represents the distance function. We use cosine similarity as the distance function in this paper.

Then we select the maximum distance d_{max} from the distance vector $\mathbf{d}$ and assign the class with the highest value as the *proto-label*². Note that each sample in the unconfident dataset also has its original label, referred to as the *noisy-label*. Based on the prototype of the *proto-label*, two circles are constructed as shown in the Figure 2: 1) **Small circle**: the distance to the prototype is less than the lower threshold α . 2) **Big circle**: the distance to the prototype is less than the upper threshold β . Based on the two circles, we are able to determine which samples should be moved from the unconfident dataset $\mathcal{D}_u$ to the confident dataset $\mathcal{D}_l$:

¹In the CIFAR-N dataset studied in this paper, the highest level of noise is approximately 40%.

²Particularly, in this paper, we perform data augmentation twice for each sample, and only when the positions of maximum similarity in the two distance vectors are the same, we construct the *proto-label* for the current sample.

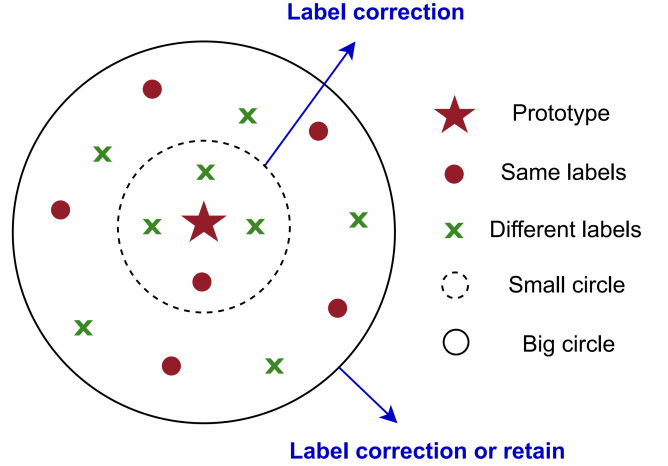


Figure 2: Noise type identifying in Proto-semi.

- **The sample that fall within the small circle ($d_{max} < \alpha$):** The sample will be assigned the *proto-label*. If the *noisy-label* is different from the *proto-label*, this indicates the factual noise is identified and corrected.
- **The samples that lie between the small and big circles ($\alpha \leq d_{max} < \beta$):** If the *noisy-label* is equal to the *proto-label*, the sample's original label, namely *noisy-label*, will be retained; Otherwise, the sample will be corrected to the *proto-label* probabilistically based on the distance to the prototype, with the probability of correction increasing as the distance decreases. This randomness allows us to correct factual noises while also utilizing semantic information from ambiguous noises.

Finally, all unconfident samples that fall within the big circle (including the small circle), after label correction or retention, will be moved to the confident dataset, creating new confident and unconfident datasets.

Semi-supervised Learning

After the new confident dataset $\mathcal{D}_l$ and unconfident dataset $\mathcal{D}_u$ are generated, inspired by the previous noisy label learning works (???), we introduce the semi-supervised training method, MixMatch (?), to enhance the training in Proto-semi. The procedure of semi-supervised learning used in Proto-semi is as follows.

We begin by augmenting the data in both the refinement confident dataset $\mathcal{D}_l$ and the unconfident dataset $\mathcal{D}_u$, resulting in the new augmented datasets $\mathcal{D}'_l$ and $\mathcal{D}'_u$, respectively. Following the approach described in (?), we apply the augmentation technique with two different strategies to each dataset:

$$\mathcal{D}'_{l1} = \text{Augment}_1(\mathcal{D}_l), \mathcal{D}'_{l2} = \text{Augment}_2(\mathcal{D}_l), \quad (6)$$

$$\mathcal{D}'_{u1} = \text{Augment}_1(\mathcal{D}_u), \mathcal{D}'_{u2} = \text{Augment}_2(\mathcal{D}_u), \quad (7)$$

where $\text{Augment}(\cdot)$ denotes the augmentation technique.

Particularly, we use the aggregated model soft outputs of $\mathcal{D}'_{u1}$ and $\mathcal{D}'_{u2}$ to guess the pseudo-labels for the augmented

unconfident dataset. After that, Mixup (?) is incorporated to combine the inputs and labels in two confident datasets (D'_{l1} , D'_{l2}) and unconfident datasets (D'_{u1} , D'_{u2}), respectively. Finally, for the combined confident dataset, we use the *cross-entropy* loss $\mathcal{L}_l$ to evaluate the difference between the outputs and true labels. While for the combined unconfident dataset, the $L2$ loss $\mathcal{L}_u$ is introduced to evaluate the discrepancy between the outputs and pseudo-labels. And the final training loss is the weighted sum of the two losses:

$$\mathcal{L}_{final} = \mathcal{L}_l + \lambda \mathcal{L}_u, \quad (8)$$

where λ is the balance factor.

We summarize the overall procedure of Proto-semi in Algorithm 1.

Algorithm 1: The Overall Procedure of Proto-semi

Input: Neural network with trainable parameters Θ , Noisy training dataset $\{(\mathbf{x}_i, \tilde{y}_i)\}_{i=1}^n$, Number of warm-up epochs T_w , proto split epochs T_p , and training epochs T_t for refining with confident examples and unconfident examples, Lower threshold α , Upper threshold β .

for $i = 1, \dots, T_w$ **do**
 Optimize network parameter Θ with Eq. (1);

for $i = 1, \dots, T_c$ **do**
 Extract confident example set $\mathcal{D}_l$ and unlabeled set $\mathcal{D}_u$ based on classifier $f(\cdot, \Theta)$ with Eq. (2) and Eq. (3);
 if $i < T_p$ **then**
 Construct prototype for each class based on $\mathcal{D}_l$ with Eq. (4);
 for $u \in \mathcal{D}_u$ **do**
 Calculate the distances between u and all class prototypes with Eq. (5);
 Select the maximum value d_{max} in all distances;
 if $d_{max} \leq \alpha$ **then**
 Assign the class corresponding to d_{max} as the label of u ;
 Move u from $\mathcal{D}_u$ to $\mathcal{D}_l$;
 if $d_{max} > \alpha$ & $d_{max} \leq \beta$ **then**
 Assign the class corresponding to d_{max} as the label of u or retain the original label of u with regard to the value of d_{max} ;
 Move u from $\mathcal{D}_u$ to $\mathcal{D}_l$;
 Training the classifier $f(\cdot, \Theta)$ with MixMatch loss on the new generated $\mathcal{D}_l$ and $\mathcal{D}_u$ with Eq. (8);
 Evaluate the obtained classifier $f(\cdot, \Theta)$.

Experiments

In this section, we evaluate the performance of Proto-semi by comparing it with representative noisy label learning

methods on the datasets that are generated from real-world annotation scenarios.

Dataset Following previous works (?), we use the CIFAR-N (?) dataset as the experimental benchmark, which has six subsets. CIFAR-N is created by relabeling CIFAR-10N/CIFAR-100N with Amazon Mechanical Turk labeling of the original CIFAR-10/CIFAR-100 dataset (?). The CIFAR-10N dataset contains five distinct noise rate options, including aggre, rand1, rand2, rand3, and worst. While the CIFAR-100N dataset only has one noise rate, namely noisy. Table 2 gives the statistics of all subsets in the CIFAR-N.

Baselines We choose the representative noisy label learning methods as the comparison methods:

- Co-Teaching (?) trains two neural networks simultaneously and lets them teach each other.
- Peer-Loss (?) introduces a new family of loss functions based on the standard empirical risk minimization framework.
- Divide-Mix (?) models the per-sample loss distribution to perform sample selection and improves MixMatch.
- ELR (?) designs a regularization term to prevent memorization of the false labels. And ELR+ (?) introduces cosine annealing learning rate to enhance training based on ELR.
- Positive-LS (?) introduces label smoothing and shows its effect is similar to label correction.
- PES-semi (?) separates the neural networks into different parts and progressively trains them.
- CAL (?) proposes a new loss function based on covariance-assisted learning.
- CORES (?) progressively sieves out corrupted examples from instance-dependent label noise. And CORES* (?) applies consistency training.
- CE (?) constructs the CIFAR-N dataset and performs the base cross-entropy model.
- Negative-LS (?) proposes a negative weight to combine the hard and soft labels.
- SOP (?) proposes a principled approach for robust training of over-parameterized deep networks.

Implementation details We implement Proto-semi by PyTorch. For fairness, all experiments are performed on a single 1080Ti GPU. We use the ResNet34 (?) as the backbone and optimize it by SGD with cosine scheduler. For each dataset, we run 5 times and report the mean and standard deviation. The learning rate, training epoch and weight decay are set to 0.02, 600, and 0.0005, respectively. And we perform a grid search to tune hyper-parameters: warm-up epoch $T_w \in \{10, 20, 30\}$, proto split epoch $T_p \in \{1, 2, 3\}$, lower threshold $\alpha \in \{0.99, 0.97, 0.95\}$, upper threshold $\beta \in \{0.94, 0.92, 0.90\}$. In semi-supervised learning, we apply random crop and flip to perform augmentation on the photos of CIFAR-N. The temperature for Mixup is set to 0.5, and the balance factor λ follows a linear decay value from the warm-up epoch to the training epoch.

Table 1: The classification results (%) over the methods on the CIFAR-N dataset w.r.t. accuracy. We highlight the best results in bold and the runner-up results with underline.

Methods	CIFAR-10N					CIFAR-100N
	aggre	rand1	rand2	rand3	worst	noisy
Co-Teaching	91.20±0.13	90.33±0.13	90.30±0.17	90.15±0.18	83.83±0.13	60.37±0.27
Peer-Loss	90.75±0.25	89.06±0.11	88.76±0.19	88.57±0.09	82.53±0.52	57.59±0.61
Divide-Mix	95.01±0.71	95.16±0.19	95.23±0.07	95.21±0.14	92.56±0.42	71.13 ± 0.48
ELR	92.38±0.64	91.46±0.38	91.61±0.16	91.41±0.44	83.58±0.13	58.94±0.92
ELR+	94.83±0.10	94.43±0.41	94.20±0.24	94.34±0.22	91.09±1.60	66.72±0.07
Positive-LS	91.57±0.07	89.80±0.28	89.35±0.33	89.82±0.14	82.76±0.53	55.84±0.48
PES-semi	94.66±0.18	95.06±0.15	95.19±0.23	95.22±0.13	92.68±0.22	<u>70.36±0.33</u>
CAL	91.97±0.32	90.93±0.31	90.75±0.30	90.74±0.22	85.36±0.16	61.73±0.42
CORES	91.23±0.11	89.66±0.32	89.91±0.45	89.79±0.50	83.60±0.53	61.15±0.73
CORES*	<u>95.25±0.09</u>	94.45±0.14	94.88±0.31	94.74±0.03	91.66±1.60	55.72±0.42
CE	87.77±0.38	85.02±0.65	86.46±1.79	85.16±0.61	77.69±1.55	55.50±0.66
Negative-LS	91.97±0.32	90.29±0.32	90.37±0.12	90.13±0.19	82.99±0.36	58.59±0.98
SOP	95.61 ± 0.13	95.28±0.13	95.31±0.10	95.39±0.11	93.24 ± 0.21	67.81±0.23
Proto-semi (Best)	95.03±0.14	95.48 ± 0.17	95.48 ± 0.21	95.67 ± 0.10	<u>92.97±0.33</u>	67.73±0.67
Proto-semi (Last)	94.82±0.15	<u>95.40±0.21</u>	<u>95.37±0.33</u>	<u>95.53±0.09</u>	92.48±0.18	67.44±0.22

Table 2: The statistics of CIFAR-N dataset

Dataset	Subset	Train/Test size	Classes	Noise rate
CIFAR-10N	aggre	50K/10K	10	9.03%
	rand1	50K/10K	10	≈18%
	rand2	50K/10K	10	≈18%
	rand3	50K/10K	10	≈18%
	worst	50K/10K	10	40.21%
CIFAR-100N	noisy	50K/10K	100	40.20%

Experimental results

Table 1 shows the evaluation results on the CIFAR-N dataset. We take the results of other baselines from the leaderboard ³ of CIFAR-N. From the table, we observe: 1) Proto-semi achieves the best average performance on all CIFAR-N datasets. Notably, Proto-semi has surpassed the state-of-the-art (SOTA) results on the rand1, rand2, and rand3 datasets. This demonstrates the effectiveness of our approach for addressing the noisy label learning problem in real annotation scenarios. 2) Proto-semi performs better on datasets with fewer classes, namely CIFAR-10N, as it can effectively construct prototype vectors for each category on these datasets. Oppositely, the generated class-prototype vectors on datasets with more categories (CIFAR-100N) are difficult to distinguish. We will discuss this phenomenon in the further experiments (See the last two experiments). 3) Sample selection-based methods, including Divide-Mix, PES-semi, and our proposed method Proto-semi all achieve relatively high performance, highlighting the importance of dividing the noisy dataset into confident dataset and unconfident dataset in noisy label learning.

³<http://competition.noisylab.com/>

Ablation Study

We further conduct ablation study to verify the importance of its main components in Proto-semi, including the repartitioning strategy in the sample selection, and the semi-supervised learning. As shown in Figure 3, we implement four variants of Proto-semi by removing the corresponding component.

We name the first variant **Proto-semi-w/o-repar**, which removes the repartitioning strategy in the sample selection. As in Figure 3, we observe declines in performance across all datasets. This suggests that our proposed repartitioning strategy can generate better confident and unconfident datasets, which ultimately helps Proto-semi achieve better classification performance. To further validate the impact of the re-partitioning strategy in detail, we conduct two ablation analyses and introduce two new variants: **Proto-semi-w/o-small** and **Proto-semi-w/o-big**. The former involves removing the construction of the small circle in the re-partitioning strategy, which means that we do not identify factual noise. On the other hand, the latter omits the construction of the large circle in the re-partitioning process, resulting in the failure to identify ambiguous noise. The results in Figure 3 show that removing either circle construction leads to a decrease in the model’s final performance compared to the complete method, indicating that both circle constructions are essential for the sample repartitioning strategy. Moreover, on most datasets, removing the small circle construction has a greater impact on performance reduction than removing the large circle construction. This suggests that constructing small circles can effectively identify factual noises and correct their labels. Notably, on the CIFAR-10(worst) dataset, retaining ambiguous noise can improve the model’s robustness and performance to some extent compared to removing the strategy that constructs small circles. The last variant, **Proto-semi-**

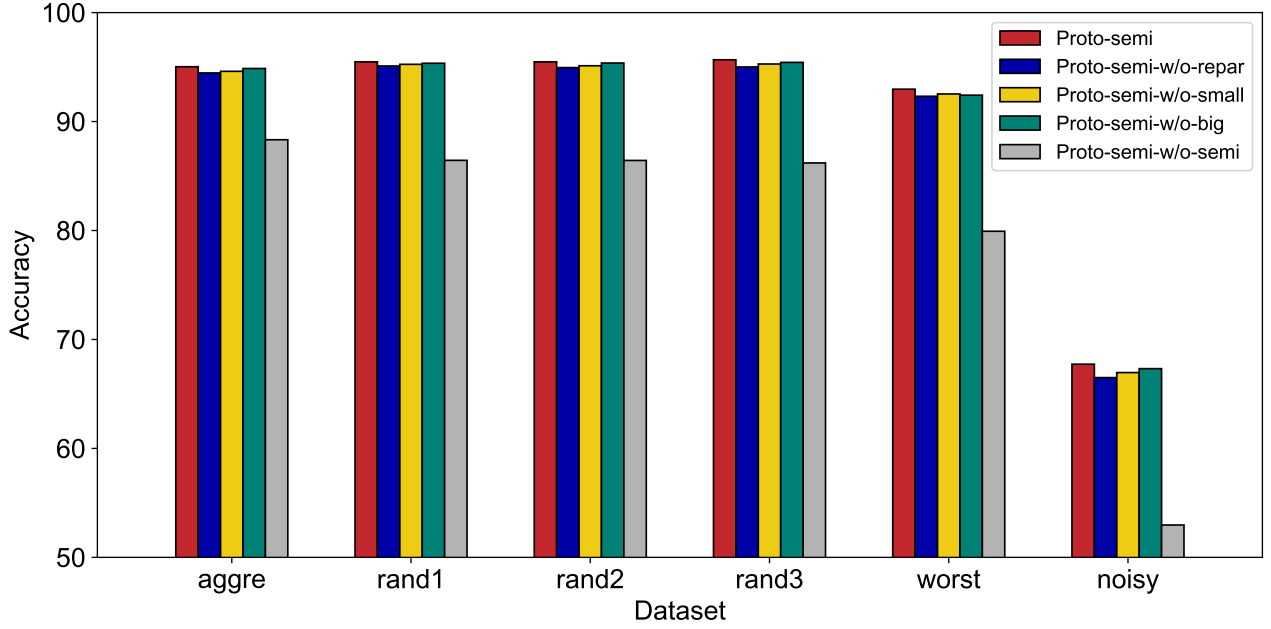


Figure 3: Ablation study on Proto-semi. From left to right: Proto-semi, Proto-semi-w/o-repar, Proto-semi-w/o-small, Proto-semi-w/o-big, Proto-semi-w/o-semi.

w/o-semi, involves discarding the semi-supervised learning step and retaining only the warmup stage described in Algorithm 1. Figure 3 shows a significant drop in performance across all datasets with this variant. Furthermore, Table 1 demonstrates that methods based on semi-supervised learning, such as PES-semi (?) and Divide-Mix (?), perform well on the CIFAR-N dataset. These findings indicate that the training augmentation provided by semi-supervised learning is highly effective for the noisy label learning in real-world annotation scenarios.

Prototype Embedding Visualization

To validate the effectiveness of prototype vector construction and its performance on learning with noisy labels, we conduct dimensionality reduction and visualization experiments using t-SNE. We choose CIFAR-10N (rand2) and CIFAR-100N (noisy) as the experimental datasets. Then a batch of 200 samples are randomly extracted from the unconfident dataset and perform visualization with all class prototype vectors. The visualization results are shown in Figure 4. Figure 4a depicts the visualization result on the CIFAR-10N (rand2) dataset. It can be observed that the 10 class prototype vectors were evenly distributed in the two-dimensional space, and the 200 sample vectors were effectively clustered around a class prototype vector. This facilitated label correction or retention based on distance. While Figure 4b shows the visualization result for the CIFAR-100N (noisy) dataset where the 100 class prototype vectors and 200 sample vectors were scattered without any discernible patterns in the two-dimensional space. Some prototype vectors even overlap, making it challenging to reassign the samples based on these class prototype vectors. Despite

using high thresholds, we found it difficult to classify these samples in our CIFAR-100N experiment.

Overall, the visualization results in Figure 4 indicate that our prototype-based repartitioning strategy can effectively learn class information in the dataset with few classes. By reassigning samples based on distances and correcting their labels, this strategy helps Proto-semi achieve good results in learning with label noise.

Label Correction Analysis

In the last experiment, we analyze the effectiveness of the proposed prototype-based repartitioning strategy in correcting labels. We record several metrics, including the size of the unconfident dataset, the number of samples falling within a small circle (See Figure 1), the number of samples corrected within the small circle, the number of samples correctly corrected, and the accuracy of correct corrections on all CIFAR-N datasets after the warm-up phase. The results are presented in Table 3, and the following discoveries can be drawn: 1) The strategy is highly effective in correcting labels. The accuracy of correction is higher than the model’s final predictive performance on almost all datasets (See Table 1). Remarkably, the strategy achieves an impressive 97.26% accuracy on the CIFAR-10N (rand3) dataset. 2) The majority of samples falling within the small circle require label correction due to factual noise. For instance, the CIFAR-10N (rand1) dataset and the CIFAR-100N (noisy) dataset have 5766 and 1823 samples within the small circle, respectively, with 5416 and 1708 samples needing correction. 3) The strategy performs better on datasets with fewer classes. This is shown by the lower number of unreliable samples, more samples within the small circle, and greater

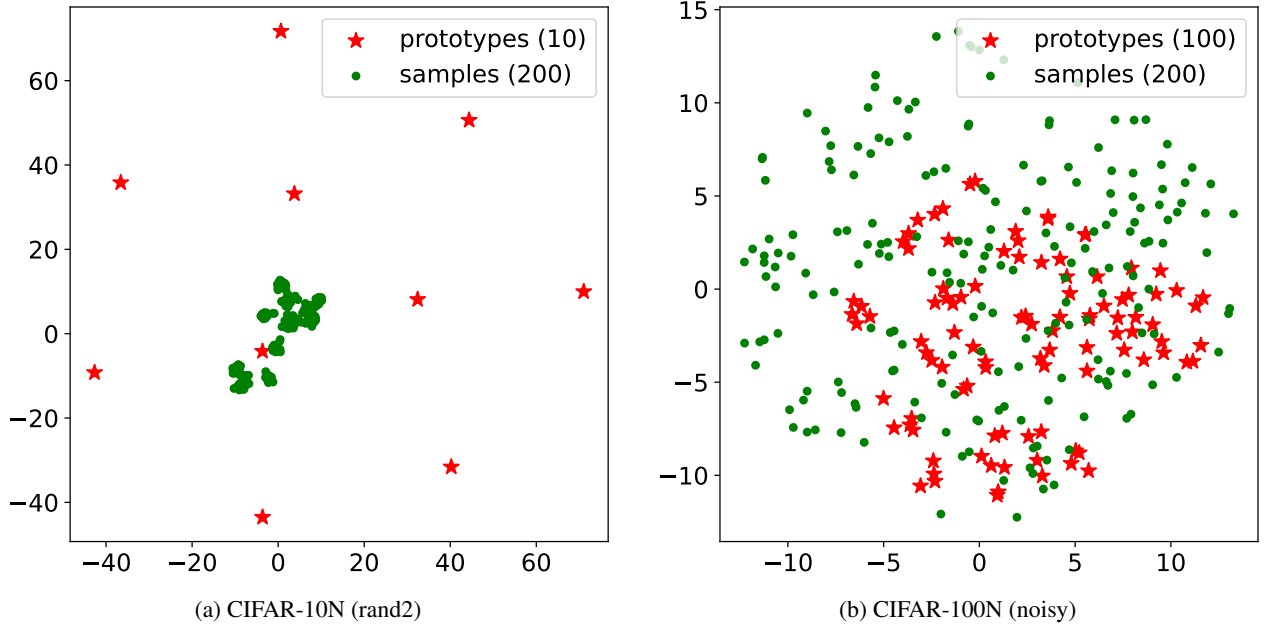


Figure 4: Visualization of prototype embedding on the CIFAR-N dataset.

Table 3: The label correction results of small circle in Proto-semi ($T_w = 20, \alpha = 0.95$).

		CIFAR-10N				CIFAR-100N
	aggre	rand1	rand2	rand3	worst	noisy
Size of unconfident dataset	10534	10995	10611	5836	21946	25096
Number of samples in small circle	4015	5766	4258	1729	9472	1823
Number of label corrections:	3992	5703	4235	1713	9381	1708
Right Number of label corrections	3846	5416	4097	1666	8723	1325
Correction accuracy	96.34%	94.97%	96.74%	97.26%	92.99%	77.58%

accuracy in label correction for all CIFAR-10N datasets compared to CIFAR-100N. These findings are consistent with the results presented in Figure 4.

Conclusion

In this paper, we present a novel sample selection-based approach, denoted as Proto-semi, for effectively identifying the type of noise and utilizing its semantics in real-world annotation scenarios. Specifically, Proto-semi divides the entire dataset into two subsets, namely confident and unconfident datasets, and subsequently leverages a distance-based technique based on prototype vectors to identify the types of annotation noise. Furthermore, the confident and unconfident datasets are adjusted based on the distances, and a semi-supervised learning strategy is incorporated to enhance the training process. Experimental results on the CIFAR-N dataset demonstrate the effectiveness of Proto-semi in addressing the problem of learning with noisy labels in real-world annotation scenarios. Avenue of future work will focus on extending our proposed method to more real-world annotation noisy datasets.

Laborum ullam similique enim nesciunt est necessitati-

bus consequuntur cupiditate eum quam voluptatibus, facilis nulla laudantium nihil voluptate suscipit, ipsum maxime ipsa consectetur natus exercitationem, libero odit ullam pariatur incidunt magnam quidem voluptas ut nostrum ipsum adipisci. Non nulla doloreque adipisci ad accusamus quae veritatis explicabo vero, blanditiis tenetur at sapiente deserunt ad delectus iste vel, quos sed libero sunt cupiditate quo voluptates fugiat reiciendis eius labore quibusdam, aliquam aperiam alias facilis voluptas ullam doloribus tenetur recusandae iure veniam. Facilis quibusdam explicabo corrupti repudiandae, tempore adipisci vel reiciendis aut quasi commodi officia quisquam beatae esse, blanditiis dicta illum nesciunt ipsam quaerat laborum, vel ullam eius esse rem quae, earum sed porro sapiente labore cumque reiciendis minima sequi dolor a alias. Praesentium quaerat quibusdam quis possimus dolor cumque, dolore veritatis dolores delectus incidunt numquam ut vel officia explicabo? Corrupti ipsam ut quos incidunt dicta recusandae fugit quo earum facilis, id ipsum illo at fugiat nemo ullam unde magni asperiores laboriosam repellendus, explicabo fuga voluptatem in. Voluptatem officiis iure aliquam doloreque consectetur reprehenderit, sed iure iusto reprehenderit sit dignissimos

praesentium vitae recusandae, error enim qui veniam alias,
iusto deserunt est aliquam ipsam nemo eos in reprehenderit
veritatis exercitationem, accusantium dicta blanditiis in.